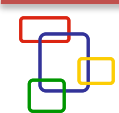
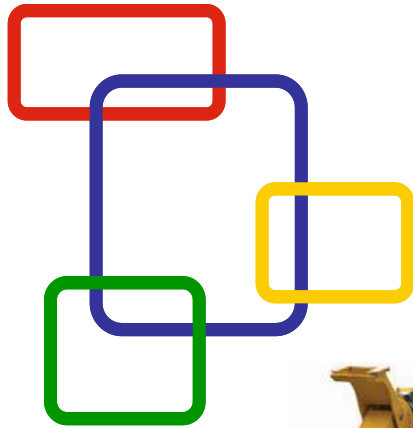


ROBOTICS

Introduction to ROBOTICS



Course Content

Introduction of robotics, Characteristics of robots, Basic requirement of robots, Laws of robotics, Advantages and disadvantages of robotics, Kinematics of robotic systems, Robotic actuators and sensors, Type of sensors, desirable features of robotic sensors, Sensing and perception, motion planning and control of robots, Computing pose of a robot, Representing position and orientation, Roll, Pitch, Yaw, Rotation matrix, Rotational transformations and parameterizations of rotations, forward kinematics, Inverse kinematics and angular velocity, Jacobian, Trajectory design and configuration space, obstacle representation and avoidance, Motion planning and control algorithms (Potential field, A^* , PRM, Voronoi diagrams, RRT, etc.), Robots programming, Practical implementation of robotics algorithms.



Reading Materials

1. **B. Siciliano, L. Sciavicco, L. Villani and G. Oriolo**, Robotics: Modelling, Planning and Control, Springer-Verlag London Limited, 2009, ISBN: 978-1-84628-6414, e-ISBN: 978-1-84628-642-1, DOI: 10.1007/978-1-84628-642-1
2. **P. Corke**, Robotics Vision and Control, 2nd Ed, Springer Tracts in Advanced Robotics, Springer International Publishing, AG, 2017.
3. **R. R. Murphy**, Introduction to AI Robotics, The MIT Press, Cambridge, Massachusetts, London, England, 2011
4. **D. J. Todd**, Fundamentals of Robot Technology, Kogan Page Ltd, 120 Pentonville Road, London, 1996
5. **X. Zhu, Y. Kim, M. A. Minor, C. Qui**, Autonomous Mobile Robots in Unknown Outdoor Environments, CRC Press Taylor and Francis Group, 2018 ISBN: 978-1-498-74055



Reading Materials

1. **B. B. K. Ayawli, R. Chellali, A. Y. Appiah, and F. Kyeremeh**, An overview of Nature Inspired, Conventional and Hybrid methods of Autonomous Vehicle Path Planning. Journal of Advanced Transportation, pp. 1-27, vol. 2018, 2018.
2. **B. B. K. Ayawli, X. Mei, M. Shen, A. Y. Appiah, and F. Kyeremeh**. Optimized RRTA* path planning method for mobile robots in partially known environment, Information Technology and Control, Vol 48 No 2, 2019, pp. 179-194 , DOI: 10.5755/j01.itc.48.2.21390
3. **B. B. K. Ayawli, X. Mei, M. Shen, A. Y. Appiah, and F. Kyeremeh**. Mobile Robot path planning in dynamic environment using Voronoi diagram and computation geometry technique. IEEE-Access, pp. 86026-86040, vol 7, 2019, DOI: 10.1109/ACCESS.2019.2925623
4. **H. Cheng**, Autonomous Intelligent Vehicles: Theory, Algorithm and Implementation, Springer London Dordrecht Heigelberg, New York, 2011, ISSN: 2191-6586, e-ISSN 2191-6594
5. **H. Yu, J. Liu, L. Liu, Z. Ju, Y. Liu and D. Zhou**, Intelligent Robotics and Applications Part IV, Springer, 2019, ISBN: 303027537X



Assessment

- Class Exercises
- Assignments
- Group Assignment and presentations
- Mid Semester Exams
- Examination



10 Marks



20 Marks



60 Marks

Total

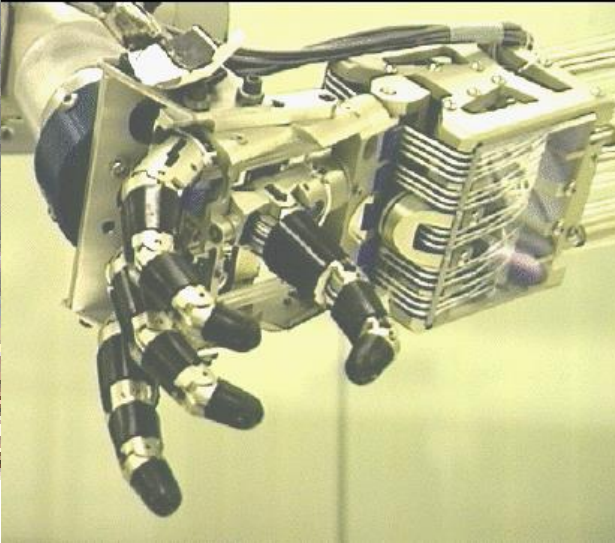
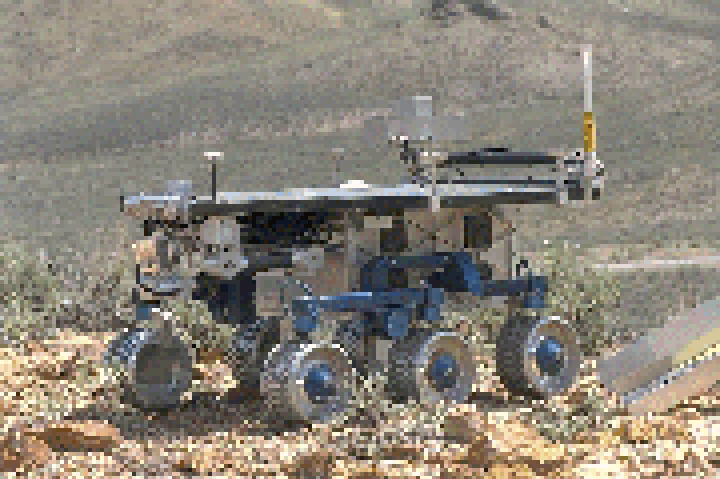


100 Marks

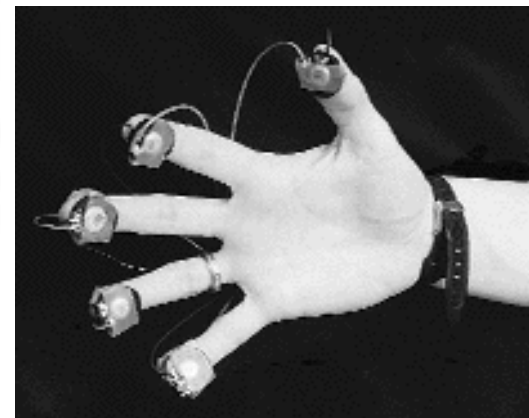
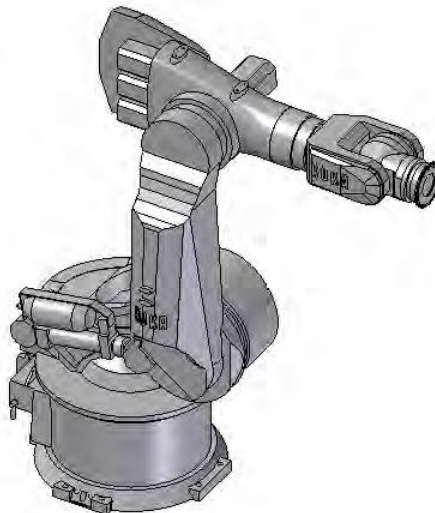
100 Marks

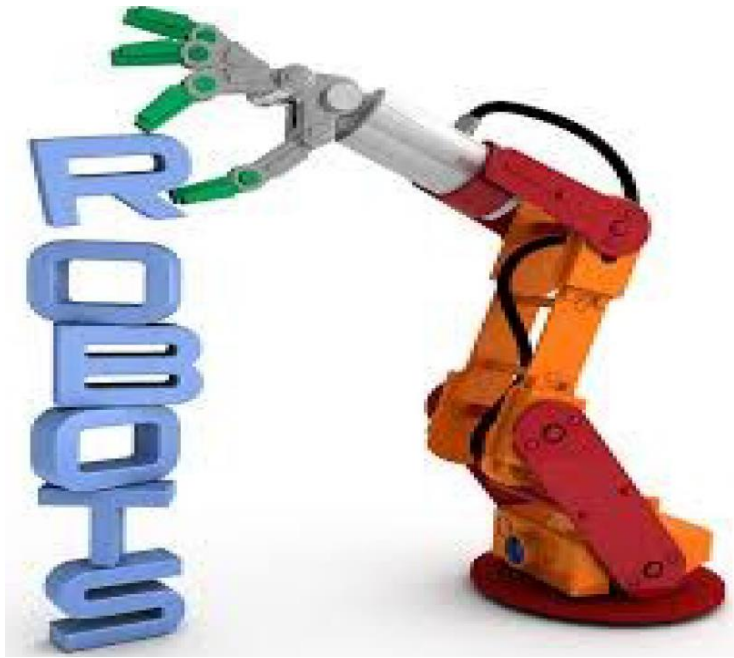
Grading system

- **85-100 - A+**
- **80 – 84 - A**
- **75 – 79 - B+**
- **70 – 74 - B**
- **65 – 69 - C+**
- **60 – 64 - C**
- **55- 59 - D+**
- **50 – 54 - D**
- **BELOW 50 - F**



Introduction to ROBOTICS





What is a robot?



What is a robot

A robot can be defined as a programmable, self controlled device consisting of electronic, electrical or mechanical units.

OR

A robot is a mechanical apparatus designed to do the work of a man. Its components are usually electromechanical and are guided by a computer program or electronic circuitry



What is a robot

- **“A robot is re-programmable, multi-functional manipulator designed to move material, parts, tools, or specialized devices through variable programmed motions for the performance of a variety of tasks.”
(Robotics Institute of America)**



The origin of the term Robot

- The term comes from a Slavic root, *robota*, with meanings associated with “**forced labor**”.
- Introduced by the **Czech** writer **Karel Čapek** in his play *R.U.R. (Rossum's Universal Robots)*, published in 1920
- The play begins in a factory that uses a chemical substitute for protoplasm to manufacture living, simplified people called *robots*.
- **Karel Čapek** himself did not coin the word. He named his brother, the painter and writer **Josef Čapek**, as its actual originator.
- He proposed *laboři* ("workers", from Latin *labor*) but the brother suggested "**roboti**"



Karel Čapek



Josef Čapek



The origin of the term Robotics

- Robotics was coined by the science fiction writer **Isaac Asimov**.
- Asimov created the "*Three Laws of Robotics*"



Basic Laws of Robotics

- Three laws proposed by **Isaac Asimov**
 1. A robot may not injure a human being or, through inaction, allow a human being to come to harm
 2. A robot must obey the orders given it by human beings except where such orders would conflict with the First Law
 3. A robot must protect its own existence as long as such protection does not conflict with the First or Second Laws



Basic requirements of Robots

- A robot typically has
 - **sensors** to sense its environment, particularly to make sure it does not hit any obstacles in its way
 - **goals** (otherwise there is no need to have the robot)
 - **planning** algorithms to determine how to accomplish those goals
 - some robots are pre-programmed with the plan steps to carry out the given goals so planning is not needed



Basic Disciplines of Robotics

- Robotics is a multi-disciplinary field. Best robotics researchers and engineers will touch upon all disciplines:
- **Mechanical Engineering** – concerned primarily with manipulator/mobile robot design, kinematics, dynamics, compliance and actuation.
- **Electrical Engineering** – concerned primarily with robot actuation, electronic interfacing to computers and sensors, and control algorithms.
- **Computer Science** – concerned primarily with robot programming, planning, and intelligent behavior.



Basic characteristics of Robots

Robots possess certain anthropomorphic characteristics- **mechanical, Energy, sensors and intelligence**

1. Robots all consist of some sort of **mechanical construction to provide movement.**

- The mechanical aspect of a robot helps it complete tasks in the environment for which it's designed.
- For example, the [Mars 2020 Rover's wheels](#) are individually motorized and made of titanium tubing that help it firmly grip the harsh terrain of the red planet.



Basic characteristics of Robots

2. Energy

- Robots need **electrical** components that **control** and **power** the machinery.
- Essentially, an electric current (a battery, for example) is needed to power a large majority of robots.



Basic characteristics of Robots

3. Sensing capability

- **sensors** to respond to inputs
- Robots need **electrical** components that sense the environment to collect data for processing for further decision making



Basic characteristics of Robots

4. Intelligence capability

- Robots contain at least some level of **computer programming which make them intelligent to make decision.**
- Without a set of code telling it what to do, a robot would just be another piece of simple machinery.
- Inserting a program into a robot gives it the ability to know when and how to carry out a task.



Sequence of operations of Robots

- The robot usually has a 3-phase sequence of operations:
 - Sense (perception),
 - Process (interpretation and planning),
 - Action (movement of some kind)



Advantages of Robots

- **They Increase Production**

- It's said that a monkey, tapping away on a typewriter forevermore, would eventually write the full works of Shakespeare.
- Humans would do it faster.
- But robots would smash it out in moments.
- They are designed for a job, and they do not stop doing it unless they breakdown or you turn them off.
- Robots can work 24/7
- Robots continue when humans have to take breaks.
- Robots keep at it when humans get sick, go on holiday, or seek a career change.
- Robots can keep working, all day every day if you want them to



Advantages of Robots

- **They are More Accurate than Humans**

- Unlike humans who make mistakes, robots are more accurate in task performance
- You calibrate their systems correctly and they do not let you down.
- They might, make mistakes if their human controller messed up first.
- Hence, Robots do not make mistakes but Humans do
- Accuracy leads to reduction of waste



Advantages of Robots (cont)

- **They are More Reliable than Humans**
 - Employees do not show up for work.
 - Friends are late for dinner dates.
 - Family-members forget to babysit your kids.
 - People have a nasty habit of being unreliable. You don't know where you stand; there's always the chance they'll let you down.
 - Robots never will. They are programmed for performance.



Advantages of Robots (cont)

- **They Help Out In Surgery**

- No matter how hard you try, how steady-handed you think you are, you inevitably end up shaking like a leaf in a gale.
- Even the best surgeons in the world aren't faultless.
- Robots are being put to good work in the operating theatre to help. They hold things in place, light things up, enter different orifices, and more.
- Mistakes are also reduced mistakes
- People recover quicker from surgery and suffer less too.



Advantages of Robots (cont)

- **They Can Do the Dirty, Boring and Dangerous Work without complain**
 - People generally lazy.
 - We do not like getting our hands dirty- especially these days. We'd rather pay someone else to do it.
 - Robots don't care. They will happily do whatever they're tasked with without complain.
 - They do the stuff we don't want to, and they will never complain about it
 - Same goes for the dangerous stuff eg. Milking of scorpion
 - Got a bomb to clear? Send the robots in. That means fewer casualties and unnecessary human injuries. Robots are handling risky business more and more in all areas of life.



Advantages of Robots (cont)

- **They are Useful in Warfare**
 - Warfare is an ugly business.
 - People die, including innocent ones.
 - Now, robots can be put to use in this regard too. That is good news for your 'team' as it were. It means fewer soldier casualties and deaths.
 - Of course, robots can also be used to do the killing. That's not quite as good if you're impartial.



Advantages of Robots (cont)

- **They Create New Jobs for People**
 - People often bemoan robots for stealing jobs.
 - In truth, though, they provide almost as many as they take. It's just that the nature of the jobs that change.
 - For instance, jobs in the coding and engineering industries are both important for designed and creating robots.



Advantages of Robots (cont)

- **They are Awesome for Space Research**
 - Robots can go to space in a way that humans can't.
 - They are on Mars. They are orbiting the planets.
They are being sent further and further into space.
 - Without them, our understanding of the universe and our place within it would be hugely reduced.
Robots are simply a fundamental part of space exploration.
 - From the production of the spacecraft and technology inside them, to the actual exploration itself, robots have a key role all-round.



Advantages of Robots (cont)

- **They are Perfect for Research in General**
 - The use of robots for research does not stop at space exploration.
 - They are used in almost every field to help in this regard. By virtue of their resilience and strength, they can go just about anywhere- to the depths of the ocean, into extreme heat or cold, and so on.
 - That makes them ideal for finding things out about things we would otherwise have no way of accessing.



Disadvantages of Robots

- **They Lead Humans to Lose Their Jobs**
 - Robots have a nasty habit of taking peoples' jobs.
 - In a capitalist system business owners have to do what it takes to maximize profits. And the brutal efficiency of robots makes them perfect for the task.
 - Humans just cannot compete with a robot that can work 24/7 without making any mistakes.
 - That fact can force people out of jobs they have done their entire lives.



Disadvantages of Robots (cont)

- **They Need Constant Power**

- Robots need oodles of electricity to run.
- That makes them expensive to run (more on this later) and potentially damaging to the environment.
- Unless we shift over to greener sources of energy, the growing demand for robots in society could lead to additional issues with global warming and greenhouse gas emissions.



Disadvantages of Robots (cont)

- **They are Restricted to their Programming**
 - Robots cannot think for themselves without programming.
 - They rely on clever humans to program them for specific tasks. And, though artificial intelligence and machine learning are coming on fast, this is a limiting factor in what Robots can do.
 - They cannot magic themselves up new capabilities at the drop of a hat.
 - They cannot think outside the box, mould themselves to novel needs, or do anything other than the task for which they're programmed.



Disadvantages of Robots (cont)

- **The Perform Relatively Few Tasks**

- In a similar way, robots are only suited, as of now, for specific roles and responsibilities.
- They come into their own in industry, research, medical practices, and the military. Outside of those domains, though, they have minimal practical usage.
- Our day to day lives are slowly becoming more robot-centric. For now, though, there's a way to go before we start putting robots to work around the house at scale.



Disadvantages of Robots (cont)

- **They Have No Emotions**

- Robots do not *feel* anything either.
- They are machines- clever chunks of metal with gears and gizmos keeping them 'alive'. But they could never feel any emotion that would allow them to empathise with, or relate to, what we're going through.
- You could never sit down and have a heart to heart with a robot.
- Sure, if the programming develops enough, then they might be able to say the right things, respond to particular cues, and react appropriately in a given situation.
- But it wouldn't be real! They would not actually be feeling anything beneath the surface.



Disadvantages of Robots (cont)

- **They Impacts Human Interaction**
 - Human interaction will suffer as robots become an increasing part of life.
 - Already, the rise of mobile phones has started this slippery slope. Just look around you in any public space and you see a mass of people staring at their screens.
 - We're more connected via the internet than ever before, but more isolated, lonely, and depressed too. We risk forgetting what it means to have an actual, real-life, human interaction.
 - There is every reason to believe that this is going to worsen as robots get an ever-greater role to play in daily life.



Disadvantages of Robots (cont)

- **They Require Expertise to Set Them Up**
 - Programming and setting up robots/computers for any given task remains something that requires particular expertise.
 - That puts the power into the hands of a limited number of skilled people around the world.



Disadvantages of Robots (cont)

- **They're Expensive to Install and Run**
 - Business owners looking to install robots in their factories/operations face significant upfront costs.
 - After all, robots aren't cheap- especially when they are high-tech, top of the line and needed for a specific task. This can put extreme financial pressure on an organisation.
 - Thanks to the electricity and manpower required to keep robots in working order, the running costs involved are high as well.
 - Another financial issue involved with robots is to do with any faults and problems that occur.
 - Everything from the materials involved to the engineers required to fix them can be expensive.



Disadvantages of Robots (cont)

- **They Pose Potential Physical Danger from attack and Malfunctions**
 - It is common for humans to work in close proximity with robots.
 - The last thing you want is for a malfunction to cause the robot to do something dangerous.
 - They might be helping out around the house, caring for peoples' wellbeing, and running any number of key tasks.
 - Imagine if there is a malfunction or somebody hacked their system and programmed them to 'misbehave'.
 - The potential repercussions could be extreme.



Disadvantages of Robots (cont)

- **It is believed they Might Take Over the World**
 - The fear of its taking over the world if care is not taken
 - The danger of robots taking over the world and killing off all the humans.
 - Needless to say, it's a terrifying prospect, made even worse by how feasible this eventuality is! Computers are getting more intelligent by the day. Sooner or later they will develop a level of sentience that enables them to make decisions for themselves.



END OF LECTURE 1

